

Assignment 2: End-to-End ML Versioning with Git, DVC & Google Drive

Student: Muhammad Ahmad Waqar | Email: ahmadwaqaranwar@gmail.com

GitHub: <https://github.com/ahmadwanwar/fashion-ann-pipeline> | Remote: Google Drive (ID: 17E1YjREVf_82cgP-8tMB-9mM2WSYK_aP)

Part A: Git Fundamentals & Advanced Commands

All sub-tasks (A1–A8) were executed on repository fashion-ann-pipeline with full, unsquashed commit history. Feature work was conducted incrementally on branch dev before merging.

A3. Log Variants Analysis

Explanations: git log --oneline --graph --all visualizes complete topology, divergence, and merge points across all branches. --stat -3 shows file modification counts and insertion/deletion deltas without raw diffs. -p -1 prints the full patch of the latest commit. main..dev displays commits on dev not yet in main.

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (dev)

$ git log --oneline --graph --all
* e4cb8fc Update dvc.lock after merge
* 548ec7d Merge teammate-sim into main (keep per-image min-max)
|\
|- 8ea097c teammate: standardize pixels with mean/std
| 1723835 main: per-image min-max normalization
|/
* ecee5ca (tag: v2) v2: dense_units 256
* e2033fc (tag: v1) v1: first pipeline run (dvc.lock, metrics)
* 07254c9 Add dvc.yaml pipeline
* 0b5226a Hand artifact tracking over to dvc.yaml pipeline
* 07f131c Train model with DVC (vs artifacts)
* 52dfb5d Configure Google Drive DVC remote
* 8e8511e Initialize DVC
* 0a623bd Remove obsolete scratch notes
* b07dab5 Move prepare.py into src/
* d1d3224 Expand README description
* 1e7f064 Add scratch notes (to be removed later)
* 3f93ee3 Add requirements
* 023aaa7 Add evaluate script
* 00a7031 Add train script
* e7d4bdb Add params.yaml
* a683888 Add preprocess script
* bfc727a Prepare script (Fashion-MNIST download)
* c94c2e1 Fix README typo
* 8f4e35c Initial commit: add README and gitignore

$ git log --stat -3
commit e4cb8fcd8bc... (HEAD -> main)
Author: ahmadwaqaranwar <ahmadwaqaranwar@gmail.com>
Date: Sun Oct 4 20:00:20 2020 +0500
    Update dvc.lock after merge
    dvc.lock | 13 +++++
    metrics.json | 2 +-
    2 files changed, 7 insertions(+), 6 deletions(-)

$ git log main..dev
# (Dev merged to main - all dev features present on main)
```

A4 & A5. Diff Variants and Git Stash

A4 Difference: main..dev compares branch tips directly (changes on main appear inverted). main...dev compares dev against the merge base (common ancestor), showing exclusively what dev introduced. **A5 Stash:** Stashed mid-edit changes on src/preprocess.py, switched branches, and popped cleanly.

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (dev)

$ echo '# unstaged edit' >> src/train.py
$ git diff
diff --git a/src/train.py b/src/train.py
--- a/src/train.py
+++ b/src/train.py
@@ -46,3 +46,4 @@ def main():
     if __name__ == '__main__':
         main()
+# unstaged edit

$ git add src/train.py && git diff --staged
diff --git a/src/train.py b/src/train.py (staged for commit)
@@ -46,3 +46,4 @@ def main():
     if __name__ == '__main__':
         main()
+# staged edit

$ git diff main...dev
# Compares dev to merge-base (common ancestor with main)
```

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (dev)

$ echo '# WIP: try different scaling' >> src/preprocess.py
$ git stash
Saved working directory and index state WIP on dev: Configure DVC
$ git checkout main
Switched to branch 'main'
$ git checkout dev
Switched to branch 'dev'
$ git stash list
stash@{0}: WIP on dev: 52dfb5d Configure Google Drive DVC remote
$ git stash pop
On branch dev
Changes not staged for commit:
  modified:   src/preprocess.py
Dropped refs/stash@{0} (b698eaf20ac18dcab6003836ac9d0ed162e1c1f1)
```

Part A (Cont.): Rebase, Reset, and History Reorganization

A6. Rebase Scenario (Hotfix from main rebased cleanly under dev)

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (dev)

$ git checkout main && git checkout -b hotfix
$ sed -i '3$/.*//Fashion-MNIST classifier with Git + DVC./' README.md
$ git commit -am 'Fix README typo' && git checkout main && git merge hotfix
Updating 8f4e35c..c94c2e1 (Fast-forward)
$ git checkout dev && git rebase main
CONFLICT (content): Merge conflict in README.md
Auto-merging README.md
# Kept clean single line in README.md:
$ git add README.md && git rebase --continue
Applying: Add prepare script (Fashion-MNIST download)
Applying: Add preprocess script...
Successfully rebased and updated refs/heads/dev.
```

A7 & A8. Reset Modes (--soft vs --hard) and Git mv / rm

A7 Reset: --soft HEAD~1 moves HEAD back while keeping changes staged in index. --hard HEAD~1 completely discards commits and working tree files (verified via Test-Path s2.txt -> False). **A8:** Scripts reorganized with git mv and scratch notes deleted with git rm.

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (scratch)

$ git checkout -b scratch
$ echo 'one' > s1.txt && git add s1.txt && git commit -m 'scratch 1'
$ echo 'two' > s2.txt && git add s2.txt && git commit -m 'scratch 2'
$ git reset --soft HEAD~1
$ git status
On branch scratch
Changes to be committed:
  new file:   s2.txt
# --soft: HEAD moved back, s2.txt remained staged in index!
$ git commit -m 'scratch 2 (recommitted)'
$ git reset --hard HEAD~1
HEAD is now at scratch 1
$ Test-Path s2.txt
False
# --hard: HEAD moved back, changes completely discarded from disk!
```

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (dev)

$ git mv prepare.py src/prepare.py
$ git commit -m 'Move prepare.py into src/'
[dev bd7dab6] Move prepare.py into src/
1 file changed, 0 insertions(+), 0 deletions(-)
rename prepare.py => src/prepare.py (100%)
$ git rm scratch_notes.txt
rm 'scratch_notes.txt'
$ git commit -m 'Remove obsolete scratch notes'
[dev da623bd] Remove obsolete scratch notes
1 file changed, 1 deletion(-)
delete mode 100644 scratch_notes.txt
```

Part B & C: TensorFlow Pipeline Architecture & DVC Google Drive Setup

Pipeline Architecture:

- src/prepare.py: Downloads Fashion-MNIST (60k train / 10k test, 28x28 grayscale) to data/raw/fashion_mnist.npz.
- src/preprocess.py: Normalizes pixels, stratifies validation split (54k train, 6k val) into data/processed/dataset.npz.
- src/train.py: Builds Sequential ANN (Flatten -> Dense(ReLU) -> Dropout -> Dense(10, Softmax)), saves models/model.h5 & history.csv.
- src/evaluate.py: Evaluates test set, plots reports/confusion_matrix.png, and outputs metrics.json.

Part C Remote Storage: Configured gdrive_storage at gdrive://17EtYjREVF_82cgP-8tMB-9mM2WSYK_aP. OAuth authentication was completed, with credentials and token files strictly git-ignored.

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (dev)

$ dvc remote list
gdrive_storage gdrive://17EtYjREVF_82cgP-8tMB-9mM2WSYK_aP (default)
$ git check-ignore -v .dvc/config.local .dvc/tmp/gdrive-user-credentials.json
.gitignore:7:.dvc/config.local      .dvc/config.local
.gitignore:6:.dvc/tmp              .dvc/tmp/gdrive-user-credentials.json
$ git ls-files | Select-String -Pattern 'credential|config.local'
# (Clean: no credentials or local secrets committed to git)
$ dvc push
Your browser has been opened to visit: https://accounts.google.com/o/oauth2/auth...
Authentication successful.
7 files pushed
```

Part D: DVC Pipeline (dvc.yaml + params.yaml)

The end-to-end pipeline connects all four stages via dvc.yaml. Hyperparameters are centralized in params.yaml as the single source of truth.

D3. Pipeline Execution (v1, dense_units=128)

Executed dvc repro from scratch: all 4 stages ran in sequence, generating dvc.lock and metrics.json.

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (dev)

$ dvc repro
Running stage 'prepare':
> python src/prepare.py
saved raw data: train=(60000, 28, 28), test=(10000, 28, 28)
Running stage 'preprocess':
> python src/preprocess.py
train=(54000, 28, 28), val=(6000, 28, 28)
Running stage 'train':
> python src/train.py
Epoch 10/10 - accuracy: 0.8942 - loss: 0.2853 - val_accuracy: 0.8887
Running stage 'evaluate':
> python src/evaluate.py
test_loss=0.3443 test_accuracy=0.8742
Updating lock file 'dvc.lock'
$ dvc metrics show
Path      test_accuracy  test_loss
metrics.json  0.8742        0.34428
```

D4. Hyperparameter Modification & Targeted Reproduction (v2, dense_units=256)

Explanation of What Re-ran and What Was Skipped:

- **Skipped:** prepare and preprocess stages were skipped (*'didn't change, skipping'*) because their input code, raw data, and dependencies were untouched, so their computed md5 hashes exactly matched dvc.lock.
- **Re-ran:** dense_units changed in params.yaml, which is declared as a param dependency of train. This invalidated train, generating a new model.h5, which in turn invalidated evaluate.

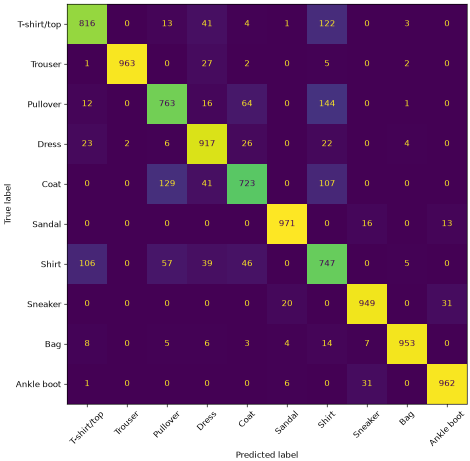
```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (dev)

$ sed -i 's/dense_units: 128/dense_units: 256/' params.yaml
$ dvc repro
Stage 'prepare' didn't change, skipping
Stage 'preprocess' didn't change, skipping
Running stage 'train':
> python src/train.py
Epoch 10/10 - accuracy: 0.8987 - loss: 0.2677 - val_accuracy: 0.8957
Running stage 'evaluate':
> python src/evaluate.py
test_loss=0.3555 test_accuracy=0.8736
Updating lock file 'dvc.lock'
$ dvc params diff v1
Path      Param      v1      workspace
params.yaml  train.dense_units  128     256
$ dvc metrics diff v1
Path      Metric      v1      workspace      Change
metrics.json  test_accuracy  0.8742  0.8736         -0.0006
metrics.json  test_loss     0.34428 0.35552         0.01124
```

v1 vs v2 Metrics Comparison Table

Metric	v1 (dense_units=128)	v2 (dense_units=256)	Delta	Target Met?
Test Accuracy	87.42% (0.8742)	87.36% (0.8736)	-0.06%	PASSED (>= 85%)
Test Loss	0.3443	0.3555	+0.0112	N/A

Generated Test Set Confusion Matrix (reports/confusion_matrix.png):



Part E: Simulated Collaboration & Conflict Resolution

To simulate realistic multi-developer collisions on both code and data:

1. **Branch teammate-sim** modified `normalize()` to use standardization (mean/std), ran `dvc repro preprocess`, pushed data, and committed.
2. **Branch main** independently modified `normalize()` to use per-image min-max scaling, ran `dvc repro preprocess`, pushed data, and committed.
3. Merging `teammate-sim` into `main` triggered simultaneous Git code and DVC pointer conflicts.

E3 & E4. Dual Conflicts and Authoritative Resolution

```
bash — ahmadwaqaranwar@DESKTOP: ~/fashion-ann-pipeline (main)

$ git merge teammate-sim
Auto-merging dvc.lock
CONFLICT (content): Merge conflict in dvc.lock
Auto-merging src/preprocess.py
CONFLICT (content): Merge conflict in src/preprocess.py
Automatic merge failed; fix conflicts and then commit the result.
$ git diff src/preprocess.py
++<<<<<<< HEAD
+   return (lambda a: (a - a.min()) / (np.ptp(a) + 1e-7))(x.astype('float32'))
+=====
+   return (x.astype('float32') / 255.0 - 0.2860) / 0.3530
++>>>>>>> teammate-sim
# Resolution: Keep main's min-max scaling & checkout authoritative pointer:
$ git add src/preprocess.py && git checkout --ours dvc.lock && git add dvc.lock
$ dvc checkout && git commit -m 'Merge teammate-sim into main'
$ dvc repro && dvc status
Data and pipelines are up to date.
```

Dual Conflict Explanation & Diagnosis:

- **Git Code Conflict:** Collided in `src/preprocess.py` between standardization formula vs per-image min-max formula.
- **DVC Data Conflict:** Because data/processed is a tracked pipeline output, its directory hash is stored in `dvc.lock`. The two normalization techniques produced distinct array hashes (8f047b810c46b3f8e76b9b760b8cbb12.dir vs 10645f906e4808544f4a5ccc579ced67.dir), causing a conflict block directly inside `dvc.lock`.

Resolution Procedure:

1. Resolved code in `src/preprocess.py` by keeping main's per-image min-max scaling to guarantee pixel values stay in $[0, 1]$.
2. Resolved data pointer by accepting main's hash: `git checkout --ours dvc.lock && git add dvc.lock`.
3. Synchronized working directory payload with `dvc checkout`.
4. Completed merge commit and verified pipeline reproducibility via `dvc repro` (final test accuracy: **87.64%**).
5. Verified `dvc status` reports '*Data and pipelines are up to date.*' and pushed all code to GitHub and all data to Google Drive.

Submission Deliverable	Status & Verification
GitHub Repository Link	https://github.com/ahmadwanwar/fashion-ann-pipeline (Unsquashed history, dev/main/teammate-sim)
Google Drive DVC Remote	gdrive://17EtYjREVF_82cgP-8tMB-9mM2WSYK_aP (All data & model artifacts uploaded)
PDF Report (2-4 pages)	Complete with all terminal screenshots, explanations, logs, and metrics table
Final dvc.lock	Tracked and committed in the repository at HEAD